Copycat

Studio 87. Free and open source (AGPL-3.0-or-later).

Intro

Copycat is a sound analyser. Play a sound through it and it estimates the note, key, chord, tempo, oscillator shape, voice count and detune, filter cutoff and slope, the envelope, and gives some EQ ideas, so you can rebuild the sound in your own synth.

Quick start

- Put Copycat on a track that plays the sound you want to study (or open the standalone app and feed it audio).
- Play the sound, then press Capture and analyse the last 2 seconds. The result appears as text in the window.
- The button turns green while sound is arriving. The audio passes through unchanged.
- Note: this version has a plain text window. The designed window (spectrogram, envelope and EQ views, Rebuild synth) is in the mockup in the repository and is not built yet.

What it reports

- Note and key, chord, and tempo of the captured sound.
- Envelope: attack, decay, sustain, release.
- Spectrum based estimates: filter cutoff and slope, oscillator shape guess, number of voices and detune.
- EQ suggestions.

Limits

- These are estimates. The cutoff is only accurate within about a factor of 2, and detune is rough. It works best on a single clean sound, not on a full mix.

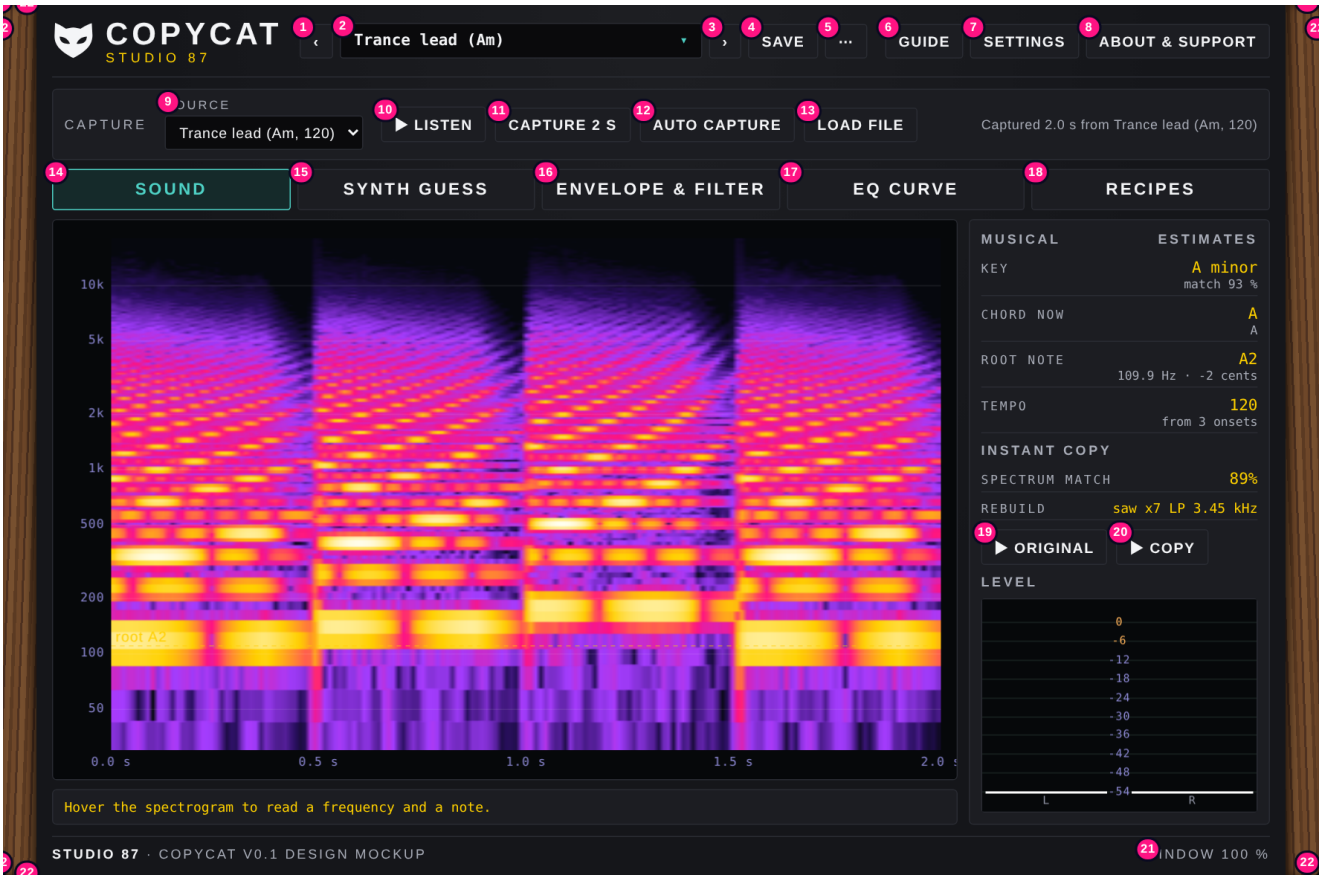
Recommendations

- Analyse one note or one chord at a time, with no reverb and no other instruments.
- Use the result as a starting point and match it by ear in your synth.
- Not tried on many real recordings yet: please report wrong results on GitHub.

Interface tour

Every pink number marks a control that has an info bubble in the interactive mockup (docs/mockup), and the tables below carry the text of those bubbles. The pictures come from the mockup, which mirrors the plugin window; the compiled plugin can differ by a few pixels of layout.

SOUND



Header bar

1	**Previous preset**	Previous preset.
2	**Init**	Open the preset list: factory presets and your own.
3	**Next preset**	Next preset.
4	**Save**	Save the current settings as your own preset (stays on this device).
5	**Delete, export or import your presets**	Delete, export or import your presets.
6	**Guide**	Quick guide: what each part does.
7	**Settings**	Info bubbles and eco mode.
8	**About & Support**	About Studio 87, privacy, and ways to support the project.

Window

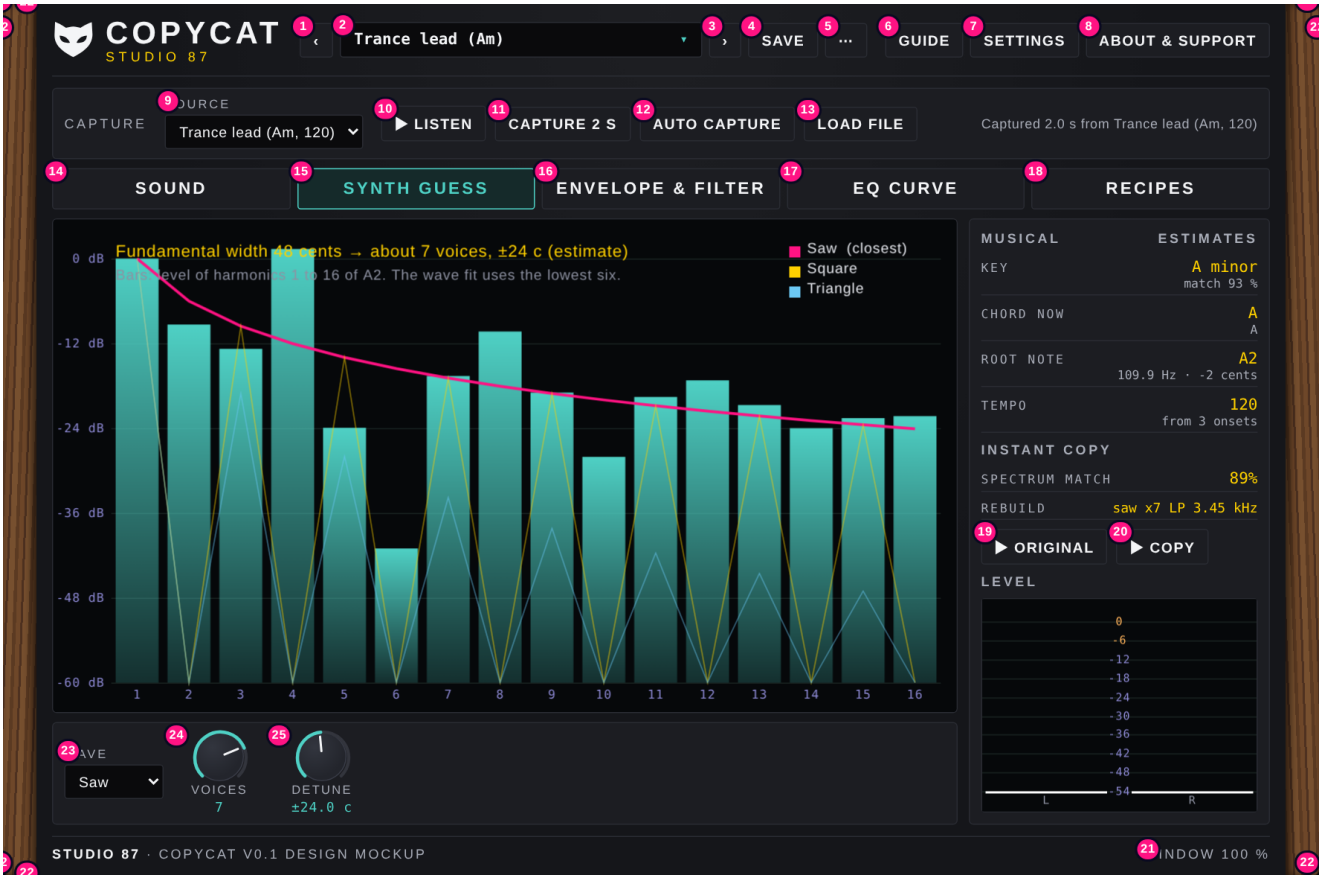
9	**Source**	What the plugin hears. In a DAW this is the track you put Copycat on. Your file uses the file you loaded.
10	**Listen**	Play the source on a loop, like a track in your DAW. Shortcut: Space.
11	**Capture 2 s**	Grab 2 seconds of what is playing now and analyze it. Shortcut: C.
12	**Auto capture**	While listening, grab a new 2 second window every time the loop turns. Off keeps the picture frozen.
13	**Load file**	Load your own audio file (wav, mp3, flac, ogg). It stays in this browser tab and is never uploaded.

14	**Sound**	Spectrogram of the captured sound: time to the right, frequency up. Hover to read a note. Shortcut: 1.
15	**Synth guess**	Harmonics of the sound against a saw, square and triangle. The closest one is the guess. Shortcut: 2.
16	**Envelope & filter**	Amp envelope with draggable points, and the filter curve. Shortcut: 3.
17	**EQ curve**	Average spectrum of the sound and an EQ you can draw to match it. Shortcut: 4.
18	**Recipes**	Written recipes to rebuild the sound in a synth. Shortcut: 5.
21	**Window 100 %**	Window size. Drag any edge or corner of the plugin to resize it; the shape stays the same. Double-click here for 100 %.
22	**Resize**	Drag to resize the window. Every edge and corner works, and the shape stays the same.

Musicalestimates

19	**Original**	Play the captured sound once.
20	**Copy**	Play the rebuilt copy once: the recipe on the Synth guess and Envelope tabs, played by the built-in synth.

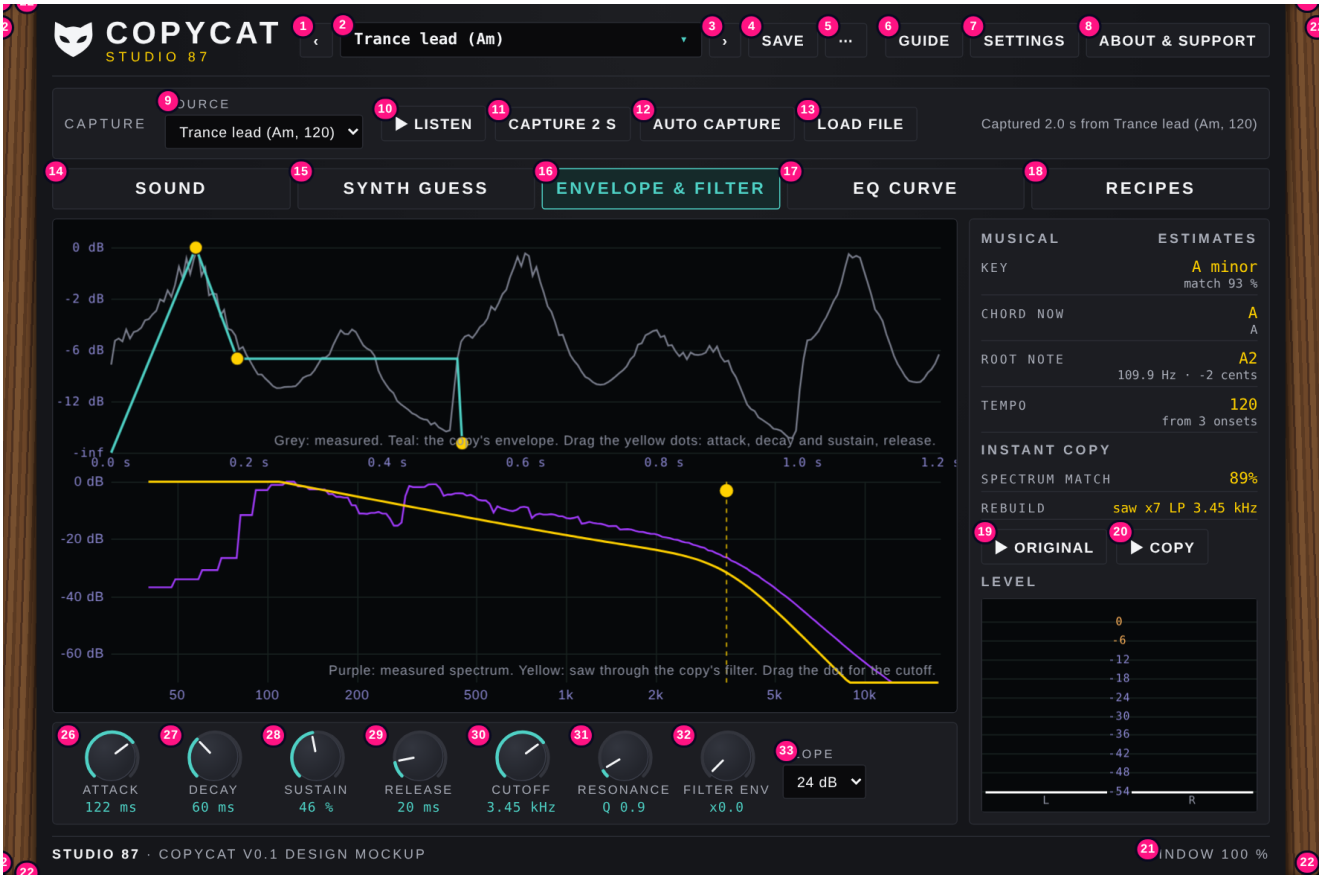
SYNTH GUESS



Window

23	**Wave**	Wave of the rebuilt copy.
24	**Voices**	How many detuned copies of the wave. The estimate comes from how wide the fundamental is. Double-click resets.
25	**Detune**	Spread of the voices in cents. Double-click resets.

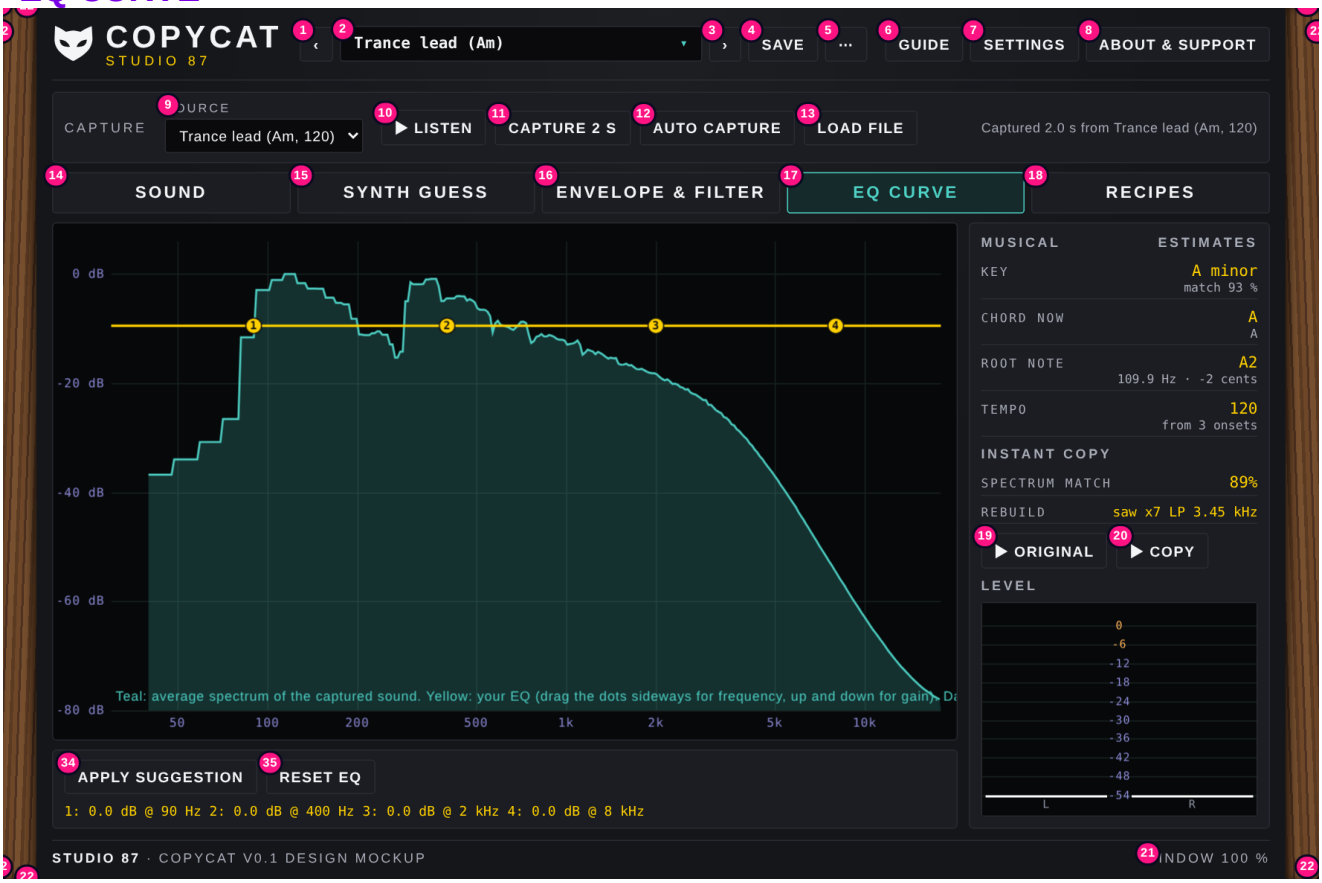
ENVELOPE & FILTER



Window

26	**Attack**	Attack of the copy. Also draggable on the Envelope display. Double-click resets.
27	**Decay**	Decay of the copy. Also draggable on the Envelope display. Double-click resets.
28	**Sustain**	Sustain of the copy. Also draggable on the Envelope display. Double-click resets.
29	**Release**	Release of the copy. Also draggable on the Envelope display. Double-click resets.
30	**Cutoff**	Low-pass cutoff of the copy. Also draggable on the filter curve. Double-click resets.
31	**Resonance**	Filter resonance. Double-click resets.
32	**Filter env**	How far the envelope opens the filter above its cutoff. Double-click resets.
33	**Slope**	Filter slope.

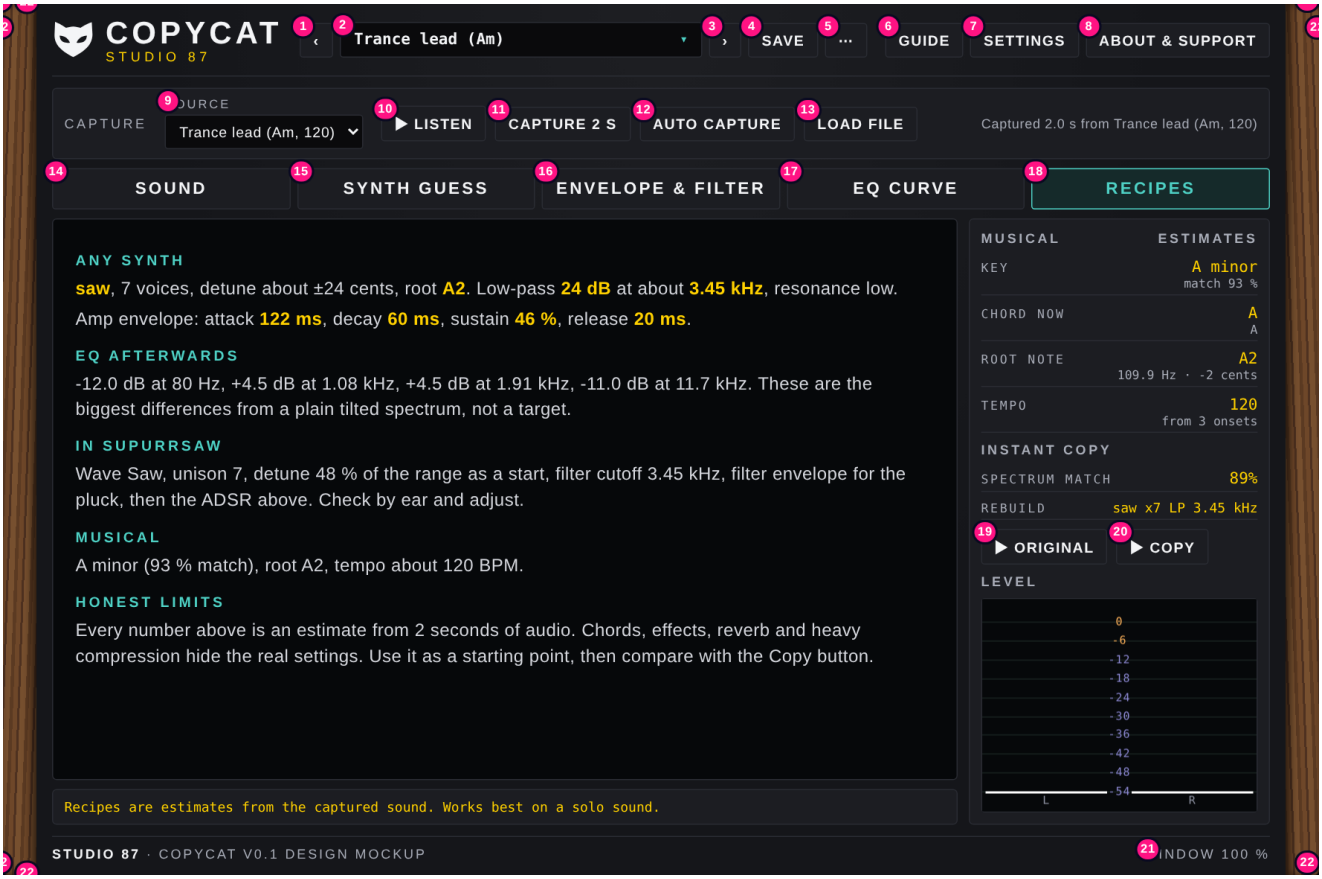
EQ CURVE



Window

34	**Apply suggestion**	Move your four EQ points to the biggest differences Copycat found.
35	**Reset EQ**	Flatten your four EQ points.

RECIPES


COPYCAT
STUDIO 87
1
‹
2
Trance lead (Am)
3
›
4
SAVE
5
···
6
GUIDE
7
SETTINGS
8
ABOUT & SUPPORT
CAPTURE
9
SOURCE
Trance lead (Am, 120)
10
▶ LISTEN
11
CAPTURE 2 S
12
AUTO CAPTURE
13
LOAD FILE
Captured 2.0 s from Trance lead (Am, 120)
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SOUND
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SYNTH GUESS
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ENVELOPE & FILTER
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RECIPES
ANY SYNTH
saw, 7 voices, detune about ±24 cents, root A2. Low-pass 24 dB at about 3.45 kHz, resonance low.
Amp envelope: attack 122 ms, decay 60 ms, sustain 46 %, release 20 ms.
EQ AFTERWARDS
-12.0 dB at 80 Hz, +4.5 dB at 1.08 kHz, +4.5 dB at 1.91 kHz, -11.0 dB at 11.7 kHz. These are the biggest differences from a plain tilted spectrum, not a target.
IN SUPURRSAW
Wave Saw, unison 7, detune 48 % of the range as a start, filter cutoff 3.45 kHz, filter envelope for the pluck, then the ADSR above. Check by ear and adjust.
MUSICAL
A minor (93 % match), root A2, tempo about 120 BPM.
HONEST LIMITS
Every number above is an estimate from 2 seconds of audio. Chords, effects, reverb and heavy compression hide the real settings. Use it as a starting point, then compare with the Copy button.
Recipes are estimates from the captured sound. Works best on a solo sound.
MUSICAL ESTIMATES
KEY
A minor
match 93 %
CHORD NOW
A
A
ROOT NOTE
A2
109.9 Hz · -2 cents
TEMPO
120
from 3 onsets
INSTANT COPY
SPECTRUM MATCH
89%
REBUILD
saw x7 LP 3.45 kHz
19
▶ ORIGINAL
20
▶ COPY
LEVEL
0
-6
-12
-18
-24
-30
-36
-42
-48
-54
L
R
STUDIO 87 · COPYCAT V0.1 DESIGN MOCKUP
21
WINDOW 100 %
22
22